

MIMIC Visualization

Lauren Escobedo
AI in Health AI 380 H
Professor Ying Ding

Visualizations - Emphasis on Socioeconomic Status and Gender

1. Percent conversion from hospital stay to ICU stay by insurance type as a socioeconomic status indicator
2. Intensity of charting/leb events during hospital stay by gender and socioeconomic status
3. Pain medication administration by race and gender
4. Rate of mental health diagnoses by gender
5. Transfer rates by ICU unit

Percent conversion from hospital stay to ICU stay by insurance type as a socioeconomic status indicator

Goal:

- Investigate if there is any apparent correlation between a patient's insurance type and whether or not they convert to an ICU admission

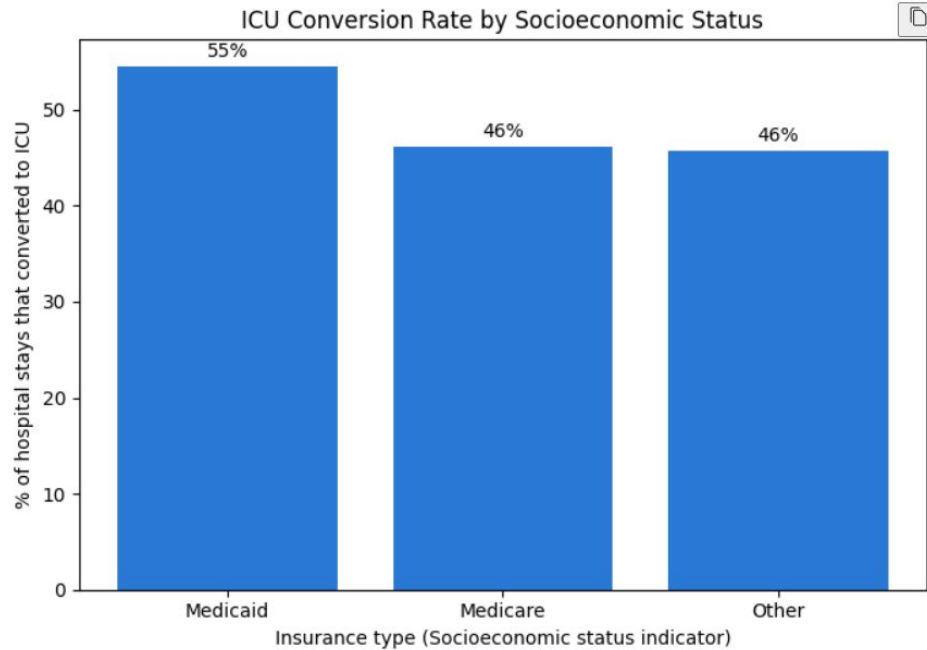
Columns needed:

- admissions: hadm_id, subject_id, insurance
- icustays: hadm_id

Method:

1. Flag each hospital admission as conversion if its hadm_id appears in icustays
2. Group by insurance type
3. Conversion rate = mean of the True/False conversion flag $\times$ 100 (by group)

Percent conversion from hospital stay to ICU stay by insurance type as a socioeconomic status indicator



There is a clear favoring for Medicaid patients to be converted to an ICU stay.

This could be due to many factors. I would continue to investigate other factors for correlation.

Intensity of charting/lab events during hospital stay by gender and socioeconomic status

Goal:

- Investigate if there is any apparent correlation between care/documentation intensity and gender/socioeconomic status

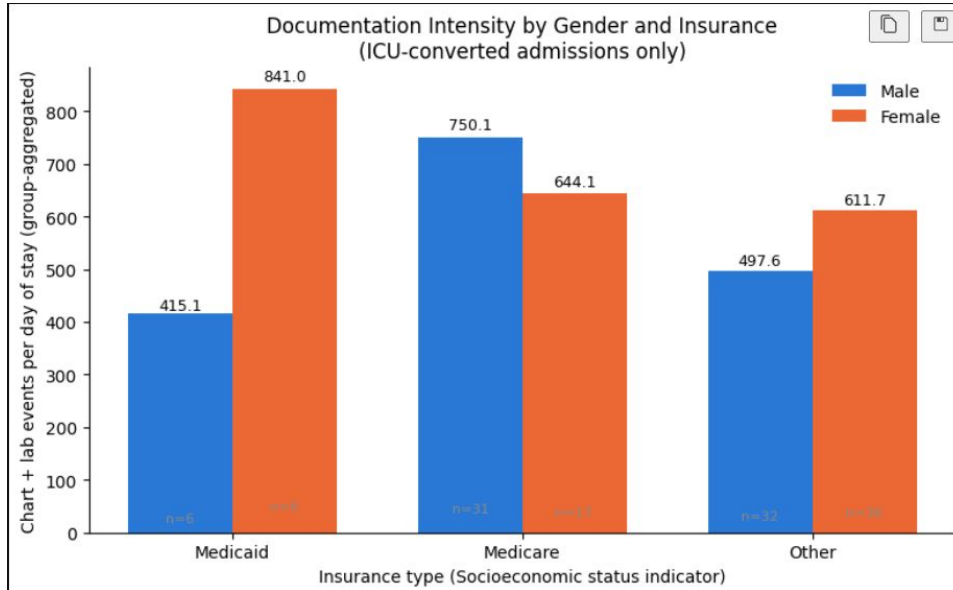
Columns needed:

- admissions: hadm_id, subject_id, insurance, admittance, dischtime
- patients: subject_id, gender
- icustays: hadm_id
- chartevents: hadm_id, charttime
- labevents: hadm_id, charttime

Method:

1. Scope to admissions that had an ICU stay (matched hadm_id in icustays)
2. Compute length of stay (LOS) in days: dischtime – admittance
3. Count chart events + lab events per admission
4. Aggregate by group: $\text{sum}(\text{events}) \div \text{sum}(\text{LOS days})$, per insurance $\times$ gender group

Intensity of charting/lab events during hospital stay by gender and socioeconomic status



The results are mostly even, except that women on Medicaid have a higher than average documentation rate, and men on Medicaid have a lower than average rate of documentation during their stay.

Pain medication administration by race and gender

Goal:

- Investigate if there is an apparent correlation between pain medication administered and a patient's race and/or gender

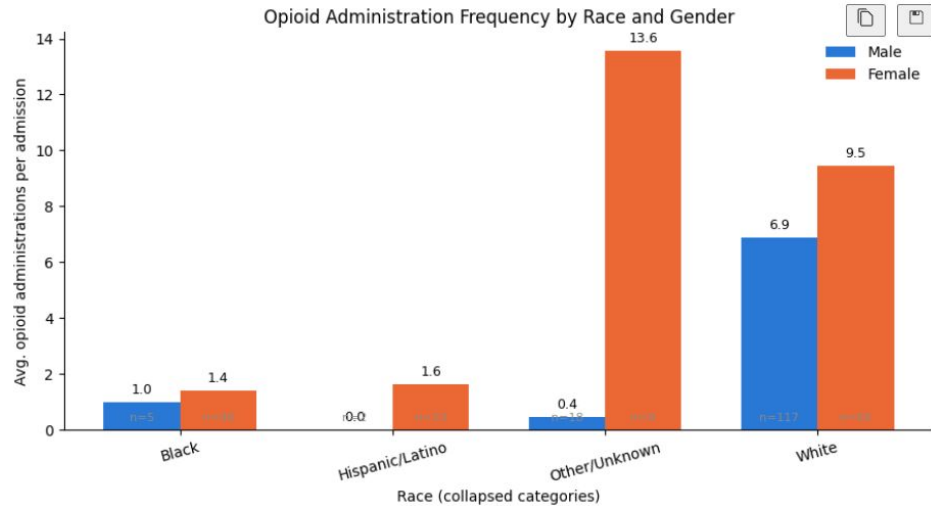
Columns needed:

- admissions: hadm_id, subject_id, race
- patients: subject_id, gender
- emar: hadm_id, medication, event_txt

Method:

1. Create a set of most common opioid names to compare against (see code for full list)
2. Match emar.medication against the defined set of common opioid names
3. Filter to rows where event_txt == 'Administered' (actual doses given, not just ordered/held)
4. Count opioid administrations per admission
5. Collapse detailed race categories into broader groups (White, Black, Hispanic/Latino, Asian, Other/Unknown)
6. Average administrations per admission, grouped by race_group × gender

Pain medication administration by race and gender



It is clear that for every category, women are administered more opioid medication. White men receive a significantly higher rate than men of other races. Hispanic men accept little to no opioid medication.

Rate of mental health diagnoses by gender

Goal:

- Investigate if there is an apparent correlation between mental health diagnosis and gender

Columns needed:

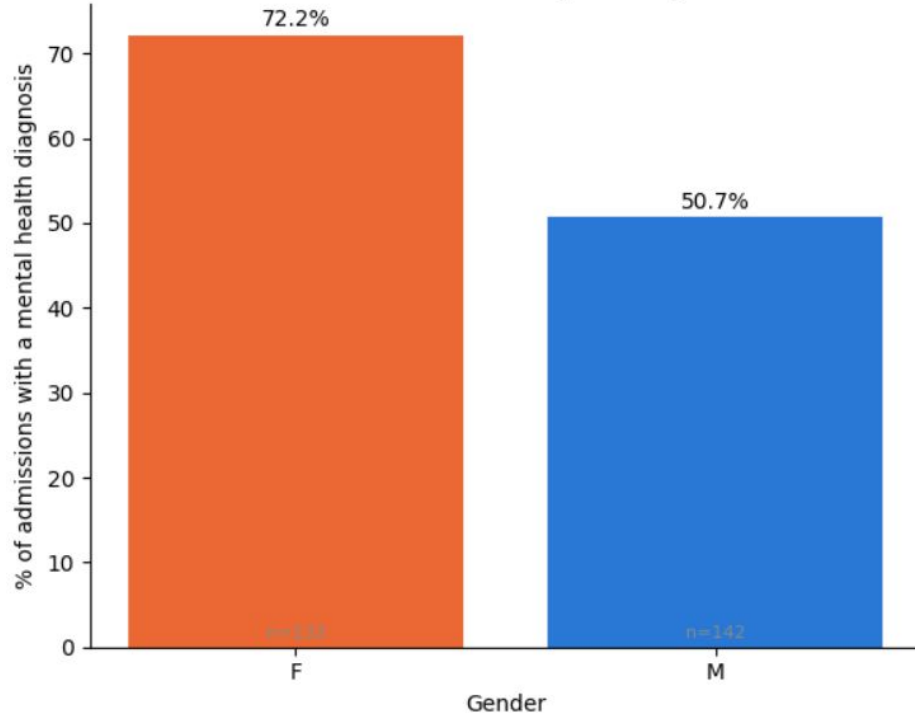
- admissions: hadm_id, subject_id
- patients: subject_id, gender
- diagnoses_icd: hadm_id, icd_code, icd_version

Method:

1. Flag mental health diagnosis rows, be aware of coding version differences:
 - a. ICD-9: code's first 3 digits fall between 290–319
 - b. ICD-10: code starts with F
2. Flag each admission as having a mental health diagnosis if its hadm_id appears among those flagged rows
3. Group by gender
4. Rate = mean of the True/False flag $\times$ 100 (per group)

Rate of mental health diagnoses by gender

Rate of Mental Health Diagnoses by Gender



We can clearly see that women are almost 25% more likely to receive a mental health diagnosis while in the hospital.

Transfer rates by ICU unit

Goal:

- Investigate the rate of patient transfers by ICU unit

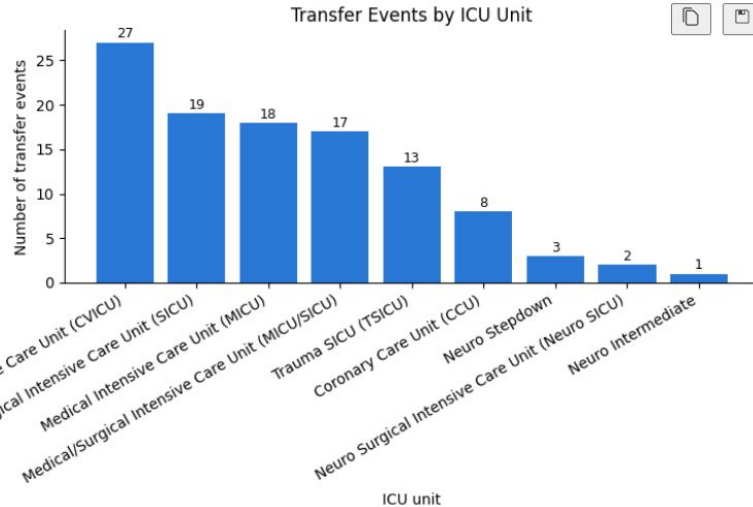
Columns needed:

- icustays: first_careunit, last_careunit
- transfers: hadm_id, eventtype, careunit

Method:

1. Build the reference set of valid ICU unit names from icustays
2. Filter transfers to rows where eventtype == 'transfer' and careunit is in that ICU unit set
3. Count transfer events per careunit
4. Sort descending by count

Transfer rates by ICU unit



These results are not shocking in any way. We see that step down and intermediary units have less transfer, while more traumatic units and surgical units transfer often. This makes sense because once a patient completes surgery, they recover in another unit.